CE334-Laboratory 6

(2025-26 II Semester)

Generating a topographic map at 1:1000 scale using LiDAR data.

Objective:

To generate a topographic map at 1:1000 scale using LiDAR data.

Methodology:

1. Classify and extract the ground using ' lasground' tool as explained in the earlier labs.
2. Generate DEM using ' las2dem' tool present in LasTools plugin.
3. Generate Contours from the DEM generated. Consider LiDAR data accuracy as 10 cm at one sigma.
    a. Load the DEM into QGIS.
    b. In menu bar, go to Raster>Extraction>Contour.
    c. Specify the input layer, Band number and interval between contour lines.
        i. Since, contour interval should be 3 times the accuracy of the map produced so what should be the contour interval for this map.
4. Generate DSM and load it into the QGIS and vectorize other features (buildings, tree location, roads) by manual tracing - in appropriate geometry as per feature in QGIS.
    a. In menu bar, go to layer>Create layer>New shapefile layer.
    b. Specify the file name, Geometry type (Point, multipoint, line, polygon) depending how a particular feature will be shown at chosen scale.
    c. Select the coordinate system based on the meta data of the provided LiDAR data.
    d. Add any attribute as a new field if you want to add any new attribute while digitization.
    e. Click OK.

f. Select the above created layer and drag it above the other layer present (DSM or the other layers created by you).

g. Right click on the created layer in which you want to add features. Select toggle editing mode.

h. In the data source manager toolbar, you can see that some icons are highlighted. Select Add Point feature or Add polygon feature or Add line feature based on the nature of feature to be plotted on this layer. The decision about a layer's geometry is taken based on application and scale of the map.

i. Then Digitize the points, polygon or line feature and provide the attribute information like house1, tree1 etc.

j. After digitizing the layer select attribute table by right clicking on the layer and view the attributes generated.

k. You can also use the following link for the tutorial if you have any doubt: https://docs.qgis.org/3.16/en/docs/training_manual/create_vector_data/create_new_vector.html

5. Use the above procedure to generate layers for all features present in the data, i.e. buildings, vegetation, roads, etc.
6. Bring all layers together to create topographic map. (https://www.qgistutorials.com/en/docs/making_a_map.html)
7. Each feature will be depicted as per the symbology of QGIS.
8. Perform other cartographic formalities required for a map - boundaries, grid, north sign, legend, graphic scale. Students can be given a sample map.
9. Report will be submission of the map in PDF.